

Open Source CFD

Tutorial #5: 3D Y-Pipe with FreeCAD

Incompressible Steady Flow: 3D Y-Pipe with simpleFOAM in FreeCAD

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Learning Objectives:

- Set up a complete CFD case using FreeCAD and CfdOF workbench
- Define boundary conditions visually in the GUI
- Generate mesh automatically with snappyHexMesh
- Run simpleFoam directly from FreeCAD
- Post-process results using ParaView integration
- Compare GUI vs terminal workflows (Tutorial #4)

This tutorial shows the work flow for the simulation of a 3D Y-shaped pipe geometry with FreeCAD and the CfdOf workbench. The geometry generated in Tutorial #4 will be reused. Instead of editing manually the dictionaries, in this case all will be done with the GUI. It is more efficient and reduces the risk of typographic mistakes. On the other side, a graphical interface, that often is not available in large computers, is needed.

Contents

1	Opening the Geometry in FreeCAD	3
2	Setting Up the CFD Analysis	4
2.1	Activating CfdOF Workbench	4
2.2	Defining the Simulation	4
3	Defining Boundary Conditions	5
3.1	Identifying Boundary Faces	5
3.2	Creating Boundary Conditions	5
4	Mesh generation with <code>snappyHexMesh</code>	7
5	Initialisation of the case and running the case	9
6	Visualisation of computation of flow rate with ParaView	11
7	Exercises	12
7.1	Flow rate in <code>inletSecondary</code>	12
7.2	Mass balance error	12

1 Opening the Geometry in FreeCAD

1. **Launch FreeCAD** from the terminal or applications menu:

2. **Open the Y-pipe geometry:**

- Go to **File** → **Open**
- Navigate to your **Tutorial #4** folder
- Select **Ypipe.FCStd**
- Click **Open**

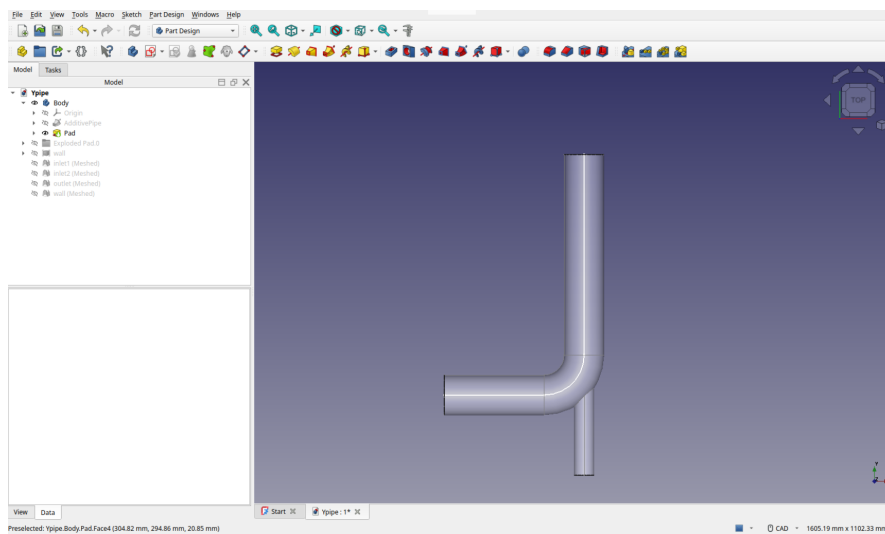


Figure 1: Opening the Y-pipe geometry

3. **Verify the geometry:**

- The model should appear in the 3D view
- The tree view should show the **Body** with all features
- Rotate and zoom to inspect all parts

TIP: If you don't have the geometry file, follow the geometry creation steps from Tutorial #4, Sections 2.1-2.2.

2 Setting Up the CFD Analysis

2.1 Activating CfdOF Workbench

1. Switch to CfdOF workbench:

- From the workbench dropdown, select **CfdOF**

2. Change output directory

- Select the menu CfdOf → Open preferences
- Change the default output directory to your folder for the Tutorial #5.

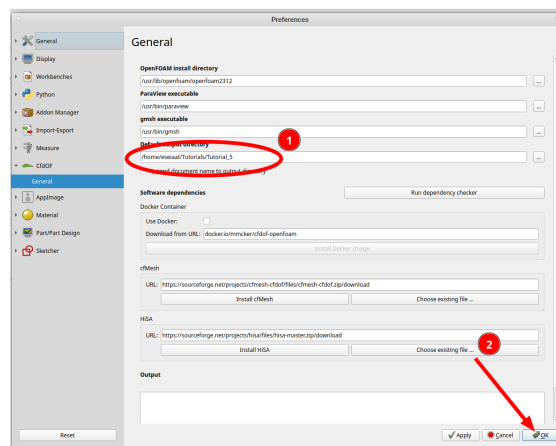


Figure 2: Selecting of output directory

3. Create a new analysis:

- Click the **Analysis container** button in the toolbar
- A new CfdAnalysis object appears in the tree

2.2 Defining the Simulation

1. Set physics models:

- Click on **Select models** button
- In the Tasks panel, set:
 - Flow type: Incompressible
 - Time: Steady
 - Turbulence model: RANS k-omega SST

- Velocity: (0 4 0) m/s (vertical direction)
- Pressure: 0 Pa (relative pressure)
- Click OK
- Turbulence specification: Intensity & Length Scale
 - Intensity (I): 5%
 - Length Scale (l): 1.4 mm
- Rename it as inletMain

TIP: CfdOf automatically calculates k and ω from these parameters using standard formulas: $k = \frac{3}{2}(UI)^2$, $\omega = \frac{\sqrt{k}}{C_\mu^{1/4}L}$ where $C_\mu = 0.09$ and L is the mixing length scale.

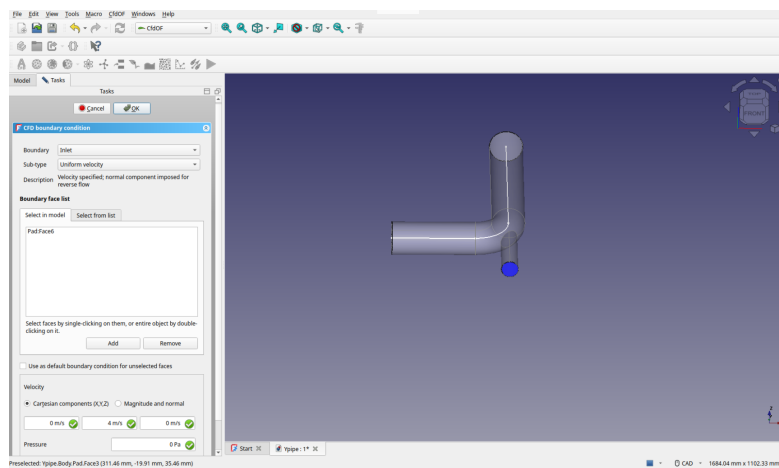


Figure 4: Setting inlet boundary conditions

2. Add inlet1:

- Repeat the same steps as with inlet2
- Type: inlet, Total pressure
- Set Pressure : 0 Pa
- Same turbulence specification as in inlet2
- Rename it as inletSecondary

3. Add outlet:

- Repeat the same steps
- Type: outlet, Static pressure

- Pressure: 0 Pa (static)
 - Leave the default name (`outlet` is not allowed because there is already an element in the tree with this name)
4. **Add walls:**
- Select all remaining faces (use Ctrl+click for multiple selection)
 - Type: wall
 - Name: walls
5. **Verify all boundaries:**
- The tree should show four boundary conditions
 - Each should have a distinct color in the 3D view

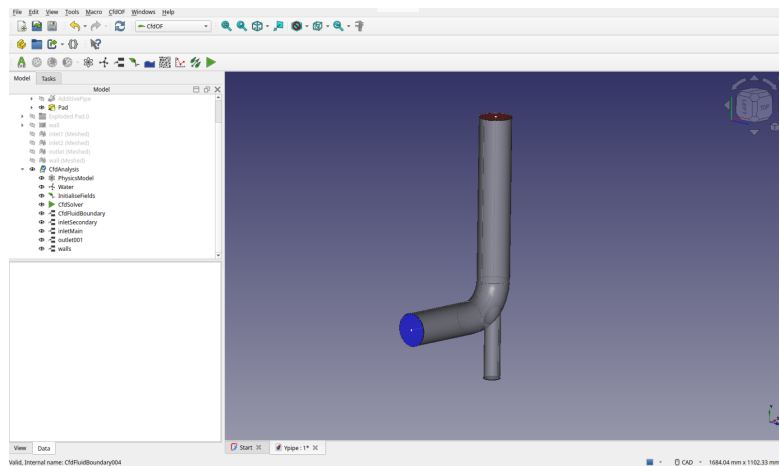


Figure 5: Completed boundary conditions

4 Mesh generation with **snappyHexMesh**

- With the Pad selected, click on **CFD Mesh**
- Choose **snappyHexMesh** as mesh utility
 - Define the **Base element size** as 5 mm
 - Click on **Search** to pick an inner point for the mesh
 - Leave the other parameters with the default values
 - Click on **Write mesh case** to write the case for meshing in the disk. Optionally the case can be reviewed

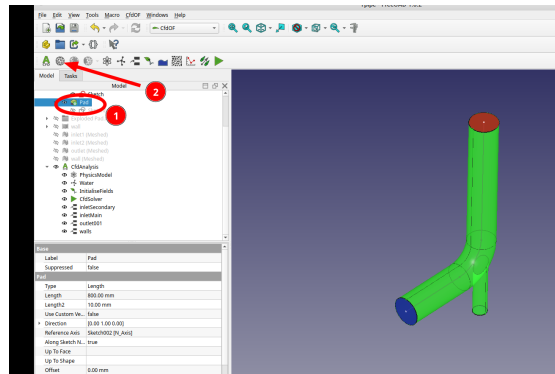


Figure 6: Adding a mesh to the model

- Click on **Run mesh**
- When the mesh is complete it can be viewed with **ParaView**. The result should be similar to the **snappy** stage of Tutorial #4.

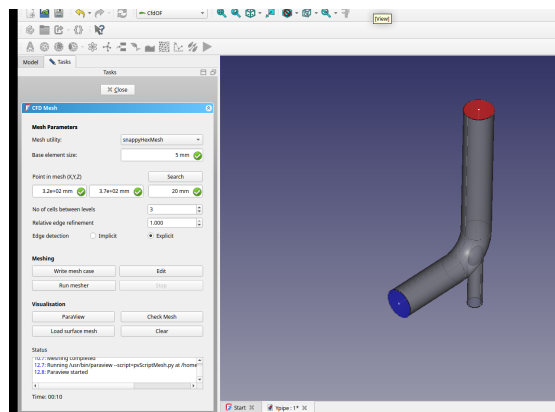


Figure 7: Setup of mesh with **snappyHexMesh**.

- To create the boundary layer, select the mesh element and click on **Mesh refinement** (see Figure 8)
 - Leave the default selection **Surface refinement**
 - Write 1.0 as **Relative mesh size**.
 - Click on the **Boundary layers** check box.
 - **Number of layers** → 3
 - **Expansion ratio** → 1.2
 - Select the **wall** element from the list (second tab on the last box). Optionally it can be selected in the model, but it is easier to select from the list.

- Save with Ok

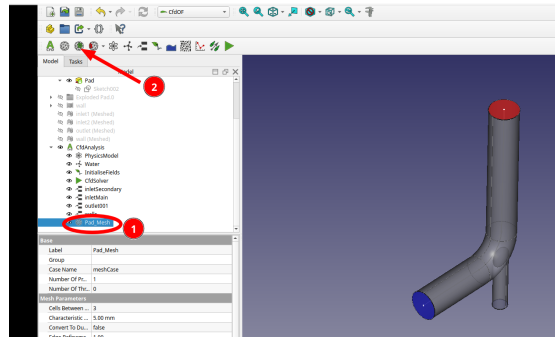


Figure 8: Add the boundary layer to the mesh

- Go back to the `Pad_Mesh` element, write again the case mesh and run the mesher to get the mesh with the boundary layer. It should be similar to the mesh generated in Tutorial #4.

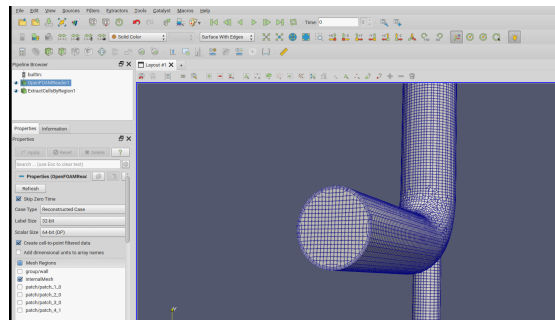


Figure 9: Final mesh viewed in ParaView

TIP: The mesh files are written to the output directory you specified in preferences. You can examine the generated `snappyHexMeshDict` to see how your GUI settings translate to OpenFOAM syntax!

5 Initialisation of the case and running the case

- Click on **Initialise** to get the initial fields for the simulation
 - * Check **Velocity** → Potential flow
 - * Check **Pressure** → Potential flow
 - * Check **Turbulence** → Use value from boundary → `inletSecondary`
 - * Save with OK

- Select the element `CfdSolver` and change the `Parallel` parameter to `false`
- Change the maximum number of iterations to 1000

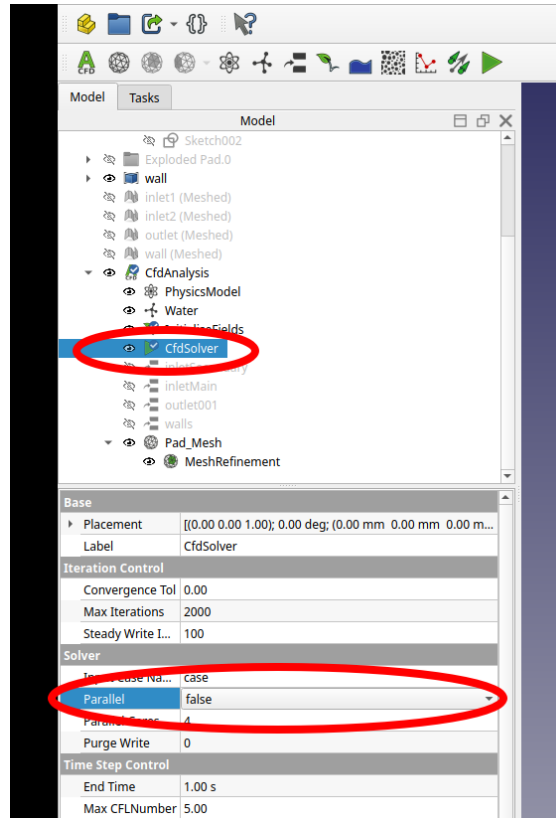


Figure 10: Disable the parallel simulation

- Double-click on `CfdSolver` and write the case.
- Run the solver. It will take about 5 minutes to complete the 1000 iterations

IMPORTANT: The residuals in Figure 11 show that convergence is not perfect. This is common for complex 3D cases with default solver settings. In practice, you would:

- * Adjust relaxation factors (try $p=0.2$, $U=0.5$)
- * Refine mesh near high-gradient areas
- * Run more iterations (2000+)

For this tutorial, we accept these residuals to keep runtime reasonable.

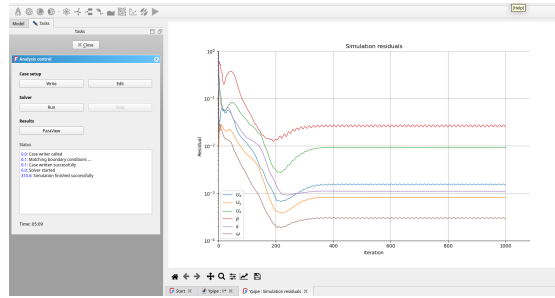


Figure 11: Residuals of the simulation

6 Visualisation of computation of flow rate with ParaView

- * Open the results with ParaView from the solver box
- * Visualisation of fields is similar to the description in Tutorial #4.
- * To compute flow rate in the `inletSecondary` patch, select uniquely this in the `OpenFOAMReader1` element

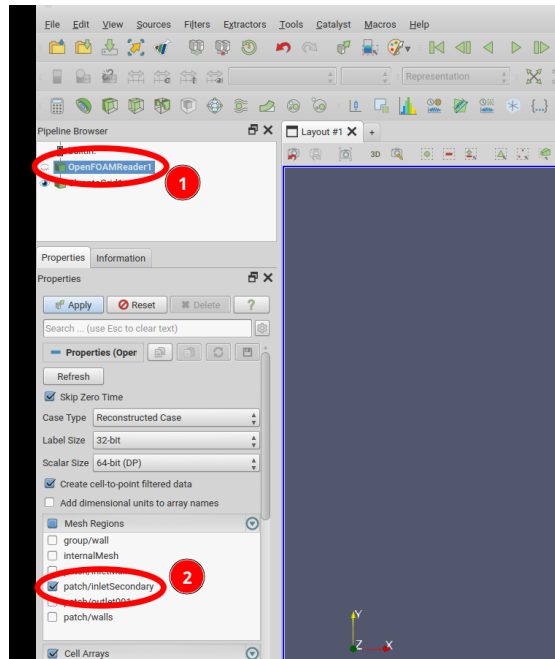


Figure 12: Selection of `inletSecondary` patch in the visualisation.

- * In the canvas, only the `inletSecondary` patch will be visible.
- * Apply the filter `Integrate Variables`, with `Cell` data as attribute.

- * The flow rate is the integral of the component x of the velocity U (see Figure 13)

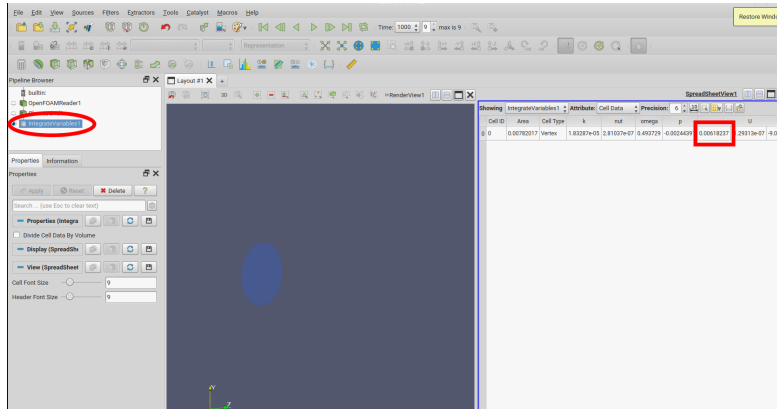


Figure 13: Computation of flow rate in `inletSecondary`

7 Exercises

The same exercises of Tutorial #4 can be done here.

7.1 Flow rate in `inletSecondary`

The flow rate obtained in this tutorial in `inletSecondary` is different from the one computed in Tutorial #4. Can you see why? How should the simulation be changed in order to get the same flow rate?

7.2 Mass balance error

Estimate the error in mass balance in the simulation,

$$\varepsilon = \frac{\|Q_o - Q_i\|}{Q_i}$$

TIP: You will need to account for many decimal places in flow rates, since this error is very small.